Shalini Mehta

Data Scientist | Data Analyst | Machine Learning Engineer
+91-9911611823 shalinimehta4485@gmail.com Portfolio Github

Education

Dr. Akhilesh Das Gupta Institute of Professional Studies **Aug 2023**
Bachelor of Technology in Artificial Intelligence and Data Science
CGPA: 8.0/10 *Delhi, India*

Technical Skills

Programming Languages: Python, SQL, Java
Libraries and Tools: NumPy, Pandas, Scikit-learn, TensorFlow, GitHub, Google Colab
Machine Learning: Regression (Linear), Classification,XGBoost, Decision Trees, Logistic Regression, Ensemble Learning, Predictive Modeling
Data & Visualization Tools: SQL, Matplotlib, Seaborn, PowerBI, Advance Excel**,** Data Visualization, Dashboard Development
Deep Learning & NLP: TensorFlow, LSTM, RNN, Transformers, NLTK, Text Classification, Sentiment Analysis
Data Science: Data Cleaning, Data Wrangling, Feature Engineering, EDA, Statistical Analysis, Model Evaluation
Databases: MySQL, PostgreSQL

Experience

DRDO | Python, XGBoost, Random Forest classifier **Jan 2026 – Feb 2026**
Data Science Intern

- Developed an end-to-end **Health & Fatigue Prediction System** using **Machine Learning** algorithms including Random Forest and **XGBoost** for multiclass fatigue classification. *Github*
- Performed **data cleaning, preprocessing, feature engineering, label encoding, outlier detection (IQR)**, and **feature scaling** on physiological and operational datasets..
- Designed predictive models to **classify fatigue levels** into Low, Moderate, and High risk categories, enabling **data-driven decision support**.
- Applied stratified **train-test splitting, hyperparameter tuning, and model optimization techniques** to improve prediction reliability and reduce overfitting.

Personal Projects

Fake News Detection System | *Python, NLP, Transformers* *Github*

- Developed an advanced multi-model pipeline integrating NLP, LSTM, and Transformer architectures achieving up to 92% accuracy.
- Implemented data wrangling using NLTK, regex, and TF-IDF for semantic feature extraction from 1000+ instances.
- Utilized ensemble strategies to boost performance, surpassing traditional classifiers like Naive Bayes and Logistic Regression by 20%.

Heart Disease Detection | *Python, Scikit-learn, Google Colab* *Github*

- Constructed a predictive health model using Logistic Regression and Random Forest classifiers to assess cardiovascular risk.
- Executed EDA with correlation matrices, box plots, and distribution curves to identify influential medical indicators.
- Trained on UCI Heart dataset with hyperparameter tuning and 10-fold cross-validation for robust evaluation.

Sentimental Analysis | *Python, NLP, Logistic Regression, LSTM, Tensorflow* *Github*

- Developed an NLP model to classify text as positive, negative, or neutral by preprocessing and embedding datasets from social media, reviews, and feedback.
- Applied machine learning and deep learning models (Logistic Regression, LSTM, Tensorflow) for accurate sentiment prediction.
- Evaluated performance using accuracy, precision, recall, and F1-score to ensure model reliability.

Certification

- **Web development(**udemy**)**
  Completed Full Stack Web Development course on Udemy, covering a wide range of modern technologies and tools, through which I developed proficiency in HTML, CSS, JavaScript, Node.js, Express.js, MySQL, and PostgreSQL.
- **CODECAP COMMUNITY (***Mentor Assistant)*
  Mentored learners on conducting Exploratory Data Analysis (EDA) with Pandas and Seaborn, uncovering insights from datasets like Iris , Facilitated project-based learning by guiding students through mini-projects involving classification tasks, sentiment analysis, and real-world data scraping.